Euclid, Book I, Proposition I.14

If Two Adjacent Angles Are Together Equal to Two Right Angles, the Outer Sides Form One Straight Line

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1 Introduction

Euclid's Proposition I.14 is the converse in spirit of the preceding Proposition I.13.

Proposition I.13 starts from a straight-line configuration and proves that the adjacent angles are together equal to two right angles. Proposition I.14 reverses this direction: if two adjacent angles are together equal to two right angles, and their outer rays lie on opposite sides of the common straight line, then those outer rays form one straight line.

In Euclid's original proof the argument is indirect. The ray opposite to one of the given rays is constructed, Proposition I.13 is applied, and equality of the resulting angles forces the constructed ray to coincide with the remaining given ray.

The Hilbert reconstruction follows the same geometric idea at a more structural level. The condition "equal to two right angles" is encoded without numerical angle measure, using an opposite ray and angle congruence. The decisive step is then supplied by the uniqueness clause of Hilbert's angle-construction axiom.

Proposition I.14 is therefore an instructive example of the interaction between Euclid's angle language, Hilbert plane separation, angle construction, and the ray calculus developed in Book Zero.

2 The Classical Configuration

Let AB be a straight line, and let two further straight lines BC and BD meet it at the common point B, with C and D lying on opposite sides of AB.

The two adjacent angles are therefore

$$\angle ABC \qquad \text{and} \qquad \angle ABD.$$

The hypothesis of Proposition I.14 is that these two angles are together equal to two right angles. The conclusion is that the two outer rays BC and BD are opposite rays and hence form one straight line.

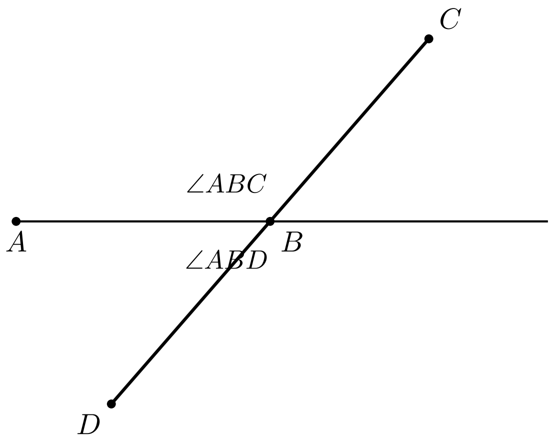

Figure 1: The classical configuration of Proposition I.14. The rays BC and BD lie on opposite sides of the straight line through A and B.

3 Statement

Theorem 1 (Euclid I.14)**.** *Let AB be a straight line, and let the straight lines BC and BD meet it at the point B on opposite sides of AB.*

If the adjacent angles

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are together equal to two right angles, then BC and BD form one straight line.

Equivalently,

$$C - B - D.$$

The proposition may therefore be viewed as a converse to the straight-line angle relation established in Proposition I.13.

The geometric content of the conclusion is stronger than mere collinearity. The point B lies between C and D, so the rays BC and BD are opposite rays with common endpoint B.

4 Classical Proof

Euclid argues by extending one of the two outer rays through the common vertex; the auxiliary configuration is shown in Figure 2.

Extend CB through B to a point E. Then

$$C - B - E,$$

so BC and BE form one straight line.

By Proposition I.13, the adjacent angles

$$\angle ABC \qquad \text{and} \qquad \angle ABE$$

are together equal to two right angles.

But by hypothesis,

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are also together equal to two right angles.

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

But if BE and BD were distinct, one of these two angles would be a proper part of the other. A proper part cannot be equal to the whole. Hence the rays BE and BD cannot be distinct.

Therefore BE coincides with BD. Since

$$C - B - E,$$

it follows that

$$C - B - D.$$

Thus BC and BD form one straight line.

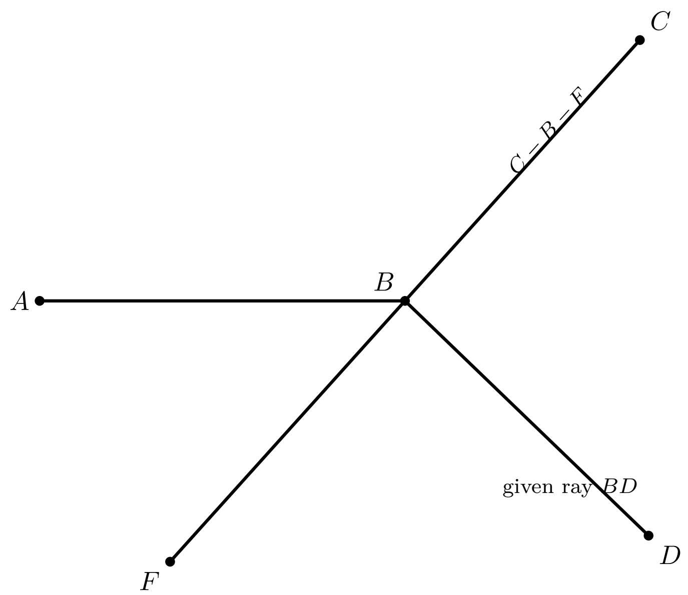

Figure 2: Wilkins auxiliary construction before the uniqueness step. The straight line CBF is the constructed extension through B; BD is the given ray. The proof shows that BD must coincide with the ray BF.

The essential structure of Euclid's argument is therefore

$$\text{extension} \longrightarrow \text{I.13} \longrightarrow \text{equality of the remaining angles} \longrightarrow \text{coincidence of the rays.}$$

The second diagram is genuinely useful, but it must be read as a *pre-coincidence* configuration. It would be misleading to draw BD as a visibly different ray and simultaneously treat $\angle ABF = \angle ABD$ as already geometrically realized. The equality is precisely what forces the two rays to coincide. Wilkins makes the missing uniqueness step explicit by using the fact that D and F lie on the same side of AB.

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction preserves the central geometric idea of Euclid's proof but reorganizes its logical dependencies.

Euclid extends CB through B to a point E and uses Proposition I.13 to obtain the supplementary angle relation. He then compares this relation with the hypothesis of Proposition I.14 and concludes that the rays BD and BE must coincide.

In the formal reconstruction, the point E is incorporated directly into the synthetic representation of the hypothesis. We require

$$C - B - E$$

together with

$$\angle ABD \cong \angle ABE.$$

Thus the intermediate appeal to Proposition I.13 is no longer needed: the supplementary configuration required by Euclid's argument is already present in the formal hypothesis.

The remaining problem is to prove that BD and BE determine the same ray from B.

At this point the reconstruction divides into two cases according to whether C, D, E are collinear.

```
Hilbert hypothesis
------------------
C-B-E
angle ABD ~= angle ABE
C and D opposite sides of base AB
        |
        v
derive: C and E opposite sides of base AB
        |
        v
case split: are C,D,E collinear?
        |
   +----+----+
   |         |
  yes        no
   |         |
line         D and E are on the same side of AB
uniqueness   |
   |         v
```

```
    |      Hilbert III.4 angle uniqueness
    |          |
    |      BE and BD are the same ray
    |          |
    +----+----+
         |
transport C-B-E along the identified ray
         |
         v
C-B-D
```

The two classical figures have different jobs. The first states the problem; the second introduces the extension used in the proof. Beyond these, another geometric copy of the same crossing is unnecessary. The formal case split is better represented as a dependency diagram, because its branches are logical alternatives rather than two prescribed Euclidean constructions.

In the collinear case, incidence and line uniqueness are sufficient.

In the noncollinear case, plane separation first places D and E on the same side of the base line. The uniqueness clause of Hilbert's angle-construction axiom then identifies the rays BD and BE. Ray calculus and transport of betweenness finally yield

$$C - B - D.$$

Schematically, the two proof architectures are

$$\begin{aligned} \text{Euclid} &: \text{extension} \to \text{I.13} \to \text{angle comparison} \to \text{ray coincidence}, \\ \text{Hilbert reconstruction} &: \text{encoded supplement} \to \text{case split} \to \text{line/angle uniqueness} \to \text{betweenness}. \end{aligned}$$

6 Hilbert Reconstruction

The Hilbert reconstruction begins by replacing Euclid's phrase "together equal to two right angles" by a purely synthetic configuration.

No numerical measure of angles and no operation of angle addition are introduced. Instead, the supplementary relation is represented by an opposite ray together with angle congruence.

6.1 Encoding the Angle Hypothesis

Choose a point E on the continuation of CB beyond B, so that

$$C - B - E.$$

The rays BC and BE are therefore opposite. Consequently, $\angle ABE$ is the angle supplementary to $\angle ABC$.

The hypothesis that

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are together equal to two right angles is represented by requiring

$$\angle ABD \cong \angle ABE.$$

The geometry is already visible in Figure 2: the witness E lies on the continuation of CB beyond B, and the formal hypothesis identifies the angle made by BD with the supplementary angle made by BE. A second drawing of the same rays would add no new information.

In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
[HilbertIncidence Geo]
(A B C D : Geo.Point) : Prop :=
exists E : Geo.Point,
Geo.Between C B E /\
Geo.AngleCongruent
A B D
A B E
```

Thus the witness E contains exactly the two pieces of geometric information needed by the reconstruction:

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

This definition should not be interpreted as a numerical definition of the sum of two angles. It is a synthetic encoding of the particular supplementary configuration occurring in Proposition I.14.

6.2 The Collinear Case

Assume first that C, D, and E are collinear.

From the definition of `HilbertAnglesEqualTwoRightAngles` we already have

$$C - B - E.$$

Hence B lies on the line determined by C and E.

Since C and D lie on opposite sides of the base line AB, the segment CD meets that line. Let X be a point of intersection, so that

$$C - X - D$$

and X lies on the base line.

No separate figure is needed for this branch. The point X is only an incidence witness: if the line through C, D, E meets the base line at X, then line uniqueness forces $X = B$. Drawing X as a second visible point near B is more confusing than informative, because the proof immediately shows that the two points coincide.

Both B and X now lie on two lines: the base line AB and the line containing C, D, E.

Suppose that

$$X \neq B.$$

By uniqueness of the line through two distinct points, the two lines would then have to coincide. In particular, C would lie on the base line.

This contradicts the hypothesis that C and D lie on opposite sides of the base line, since a point belonging to the line itself cannot be one of the two opposite-side points.

Therefore

$$X = B.$$

Substituting this equality into

$$C - X - D$$

gives

$$C - B - D.$$

Thus the collinear branch requires no angle argument at all. Once C, D, E are known to be collinear, the conclusion follows entirely from betweenness, plane separation, incidence, and uniqueness of the line through two distinct points.

6.3 The Noncollinear Case

Assume now that C, D, and E are not collinear.

The segment CD meets the base line because C and D lie on opposite sides of it. The segment CE also meets the base line, namely at B, since

$$C - B - E.$$

Because C, D, E form a genuine triangle, the Hilbert plane-separation theory implies that the remaining endpoints D and E lie on the same side of the base line; this is the configuration shown in Figure 3. Thus

$$\mathrm{SameSide}(D, E; AB).$$

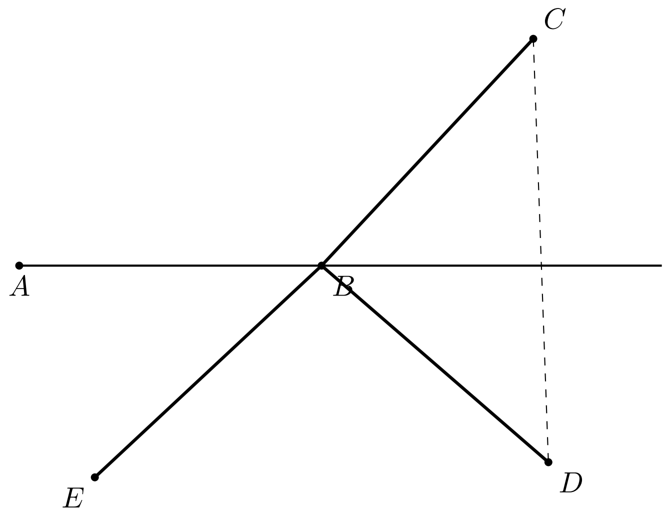

Figure 3: Noncollinear branch of the Hilbert reconstruction. The relation $C - B - E$ puts E opposite C across B; plane separation places D and E on the same side of the base line AB. The equality of the two angles then invokes the uniqueness clause of Hilbert III.4.

We now use Hilbert's angle-construction axiom.

The angle $\angle ABE$ is nondegenerate. Construct, on the side of the base line containing D, a ray BZ satisfying

$$\angle ABZ \cong \angle ABE.$$

Hilbert III.4 includes a uniqueness clause: on the prescribed side of the base ray there is only one ray realizing the prescribed angle.

Since E lies on the same side of the base line as D, and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\mathrm{SameRay}(B, Z, E).$$

On the other hand, the encoded hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

and hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

The point D lies on the side on which BZ was constructed. Applying the same uniqueness clause again gives

$$\mathrm{SameRay}(B, Z, D).$$

We therefore have two same-ray relations with the common reference ray BZ:

$$\mathrm{SameRay}(B, Z, E), \qquad \mathrm{SameRay}(B, Z, D).$$

The Book Zero ray calculus now yields

$$\mathrm{SameRay}(B, E, D).$$

Finally, from the encoded supplementary configuration we already know

$$C - B - E.$$

Replacing E by another point D on the same ray from B preserves the betweenness relation. Transport of betweenness along equal rays therefore gives

$$C - B - D.$$

This completes the noncollinear branch and hence the Hilbert reconstruction of Proposition I.14.

7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_14
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase :
HilbertIncidence.OnLine A base)
(hBbase :
HilbertIncidence.OnLine B base)
(hOppCD :
HilbertOppositeSide Geo C D base)
(hTwoRight :
HilbertAnglesEqualTwoRightAngles
Geo A B C D) :
Geo.Between C B D := by
```

The formal conclusion

$$\mathrm{Between}(C, B, D)$$

expresses directly that B lies between C and D, and hence that the rays BC and BD form one straight line.

The principal dependency structure of the proof is

```
HilbertAnglesEqualTwoRightAngles
        |
        +-- C-B-E
        +-- angle ABD ~= angle ABE
        |
HilbertOppositeSide(C,D; base AB)
        |
case split on Collinear(C,D,E)
        |
   +----+--------------------+
   |                         |
collinear                 noncollinear
   |                         |
line uniqueness           SameSide(D,E; base AB)
   |                         |
X = B                    Hilbert III.4 uniqueness
   |                         |
   |                     same ray BE, BD
   |                         |
   +-----------+-------------+
               |
hilbert_between_transport_sameRays
               |
               v
             C-B-D
```

The proof introduces no new geometric axiom. The new predicate

```
HilbertAnglesEqualTwoRightAngles
```

serves only as an interface representation of the angle hypothesis. The actual derivation uses the existing incidence, order, plane separation, angle-construction, and ray machinery.

A notable difference from Euclid's dependency graph is that the Lean theorem does not invoke Proposition I.13. The supplementary configuration needed in the classical proof has already been encoded in the hypothesis, and the identification of the relevant rays is obtained directly from Hilbert III.4.

8 Relationship with Earlier Propositions

8.1 Proposition I.13

Proposition I.13 gives the direct statement: when one straight line stands on another, the two adjacent angles are either two right angles or together equal to two right angles.

Proposition I.14 is the converse. Starting from the supplementary-angle condition, it proves that the two outer rays form one straight line.

Thus I.13 and I.14 form a natural direct/converse pair.

8.2 Proposition I.4

The reconstruction ultimately relies on the Hilbert congruence infrastructure whose triangle-level consequences include SAS. More importantly for I.14, Hilbert III.4 supplies uniqueness of angle construction on a prescribed side of a ray.

This uniqueness principle plays the role of a rigidity theorem: once two rays on the same side make congruent angles with the same initial ray, they must coincide.

8.3 Transition to Proposition I.15

Propositions I.13 and I.14 establish the supplementary and straight-line geometry surrounding an intersection. Proposition I.15 then turns to the opposite angles produced by two intersecting lines and proves their congruence.

The three propositions therefore form a coherent local block: straight-angle structure, its converse, and vertical angles.

9 Hilbert and Book Zero Components Used

The proof of Proposition I.14 uses several previously established components of the Hilbert and Book Zero layers. Their roles in the reconstruction are recorded here explicitly.

9.1 `hilbert_third_side_endpoints_sameSide`

In the noncollinear branch, the points C, D, E form a genuine triangle. The base line meets the interior of CD, because C and D lie on opposite sides of it, and it meets the interior of CE at B, since

$$C - B - E.$$

The theorem

```
hilbert_third_side_endpoints_sameSide
```

then shows that the remaining endpoints D and E lie on the same side of the base line.

Schematically,

$$\begin{matrix} C - X - D, \\ C - B - E, \\ X, B \in AB \end{matrix} \qquad \Longrightarrow \qquad \text{SameSide}(D, E; AB).$$

This is the plane-separation step that makes the uniqueness clause of Hilbert's angle-construction axiom applicable to both D and E.

9.2 Hilbert III.4: Uniqueness of Angle Construction

Hilbert's angle-construction axiom provides not only the existence of a copy of a given angle but also uniqueness on a prescribed side of the base ray.

In Proposition I.14 a ray BZ is constructed so that

$$\angle ABZ \cong \angle ABE.$$

Both E and D lie on the prescribed side of the base line. The uniqueness clause can therefore be applied twice, yielding

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D).$$

This is the formal replacement for Euclid's final inference that equal angles on the same side determine the same straight line.

9.3 `bookZero_36_ray3`

The Book Zero theorem

```
bookZero_36_ray3
```

is a ray-composition principle.

In the present proof,

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D)$$

imply

$$\mathrm{SameRay}(B, E, D).$$

Thus the auxiliary ray BZ introduced by Hilbert III.4 can be eliminated, leaving the geometrically relevant statement that BE and BD are the same ray.

9.4 **`hilbert_between_transport_sameRays`**

The final step uses

```
hilbert_between_transport_sameRays
```

to transport strict betweenness along corresponding rays.

We know

$$C - B - E$$

and have established

$$\mathrm{SameRay}(B, E, D).$$

The ray from B toward C is unchanged, while E is replaced by the point D on the same ray from B. Therefore the theorem gives

$$C - B - D.$$

This is precisely the formal conclusion of Proposition I.14.

The dependency path of the noncollinear branch can therefore be summarized as

$$\begin{gathered}\text{plane separation}\\ \downarrow\\ \mathrm{SameSide}(D, E; AB)\\ \downarrow\\ \text{Hilbert III.4}\\ \downarrow\\ \mathrm{SameRay}(B, Z, E), \mathrm{SameRay}(B, Z, D)\\ \downarrow\\ \texttt{bookZero\_36\_ray3}\\ \downarrow\\ \mathrm{SameRay}(B, E, D)\\ \downarrow\\ \texttt{hilbert\_between\_transport\_sameRays}\\ \downarrow\\ C - B - D.\end{gathered}$$

10 Formalization Notes

Proposition I.14 is a good example of a classical converse whose formal proof is governed by uniqueness rather than by numerical angle arithmetic.

The hypothesis says that two adjacent angles have the same total effect as a straight angle. In a traditional proof this is often read directly from the diagram. In the Hilbert reconstruction, the decisive step is to compare rays using the uniqueness clause of angle construction.

The proof naturally separates into two cases.

If the relevant points are already collinear, the desired straight-line conclusion follows from order and collinearity reasoning.

If they are not collinear, the angle hypothesis determines a ray that would have to occupy the same side and form the same angle as another already specified ray. Hilbert III.4 then forces the two rays to coincide. The resulting same-ray information is transported back to betweenness, contradicting the assumed noncollinear configuration.

Schematically,

$$\text{supplementary-angle hypothesis} \longrightarrow \text{same-side information} \longrightarrow \text{angle-construction uniqueness} \longrightarrow \text{same ray} \longrightarrow$$

This is structurally different from simply writing an equation such as $\alpha + \beta = 180°$. No numerical angle measure is introduced. The straight-line conclusion is recovered entirely from incidence, order, angle congruence, side-of-line information, and uniqueness.

11 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}
\text{adjacent-angle hypothesis} \\
\downarrow \\
\text{collinear / noncollinear case split} \\
\downarrow \\
\texttt{hilbert_third_side_endpoints_sameSide} \\
\downarrow \\
\text{Hilbert III.4 uniqueness} \\
\downarrow \\
\texttt{bookZero_36_ray3} \\
\downarrow \\
\texttt{hilbert_between_transport_sameRays} \\
\downarrow \\
\text{required straight-line configuration} \\
\downarrow \\
\texttt{euclid_proposition_14}.
\end{array}$$

12 Conclusion

Proposition I.14 completes the converse half of the supplementary-angle theory begun in I.13.

Its formal reconstruction is especially instructive because it avoids introducing angle addition or numerical angle measures. The proof instead uses Hilbert's side-of-line and angle-construction uniqueness machinery to show that the geometry of the rays is rigid enough to force a straight line.

The proposition therefore illustrates a recurring feature of the project: statements that look algebraic in modern notation can often be recovered synthetically from incidence, order, congruence, and uniqueness.

Together with I.13, it prepares the local theory of intersecting lines used immediately in Proposition I.15.